

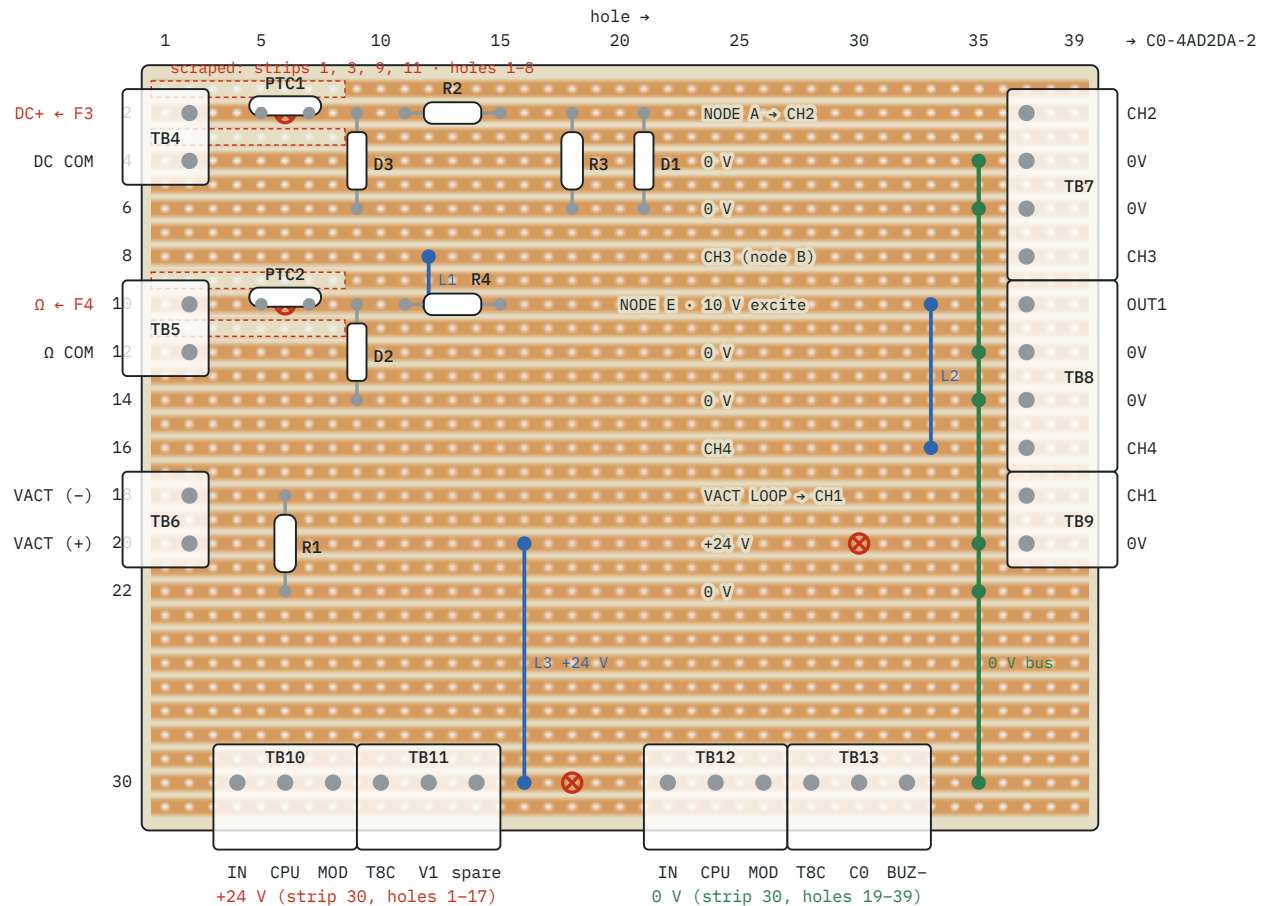
Bench Meter Stripboard

All the low-voltage conditioning (DC volts, ohms, the VACT loop resistor) and 24 V distribution on one ST2 stripboard, from the input jacks' fuses to the C0-4AD2DA-2 terminals. Positions are given as **strip** / **hole**: strips 1–31 run top to bottom, holes 1–39 left to right, viewed from the component side.

120 VAC stays off this board

Stripboard strips are only about 1 mm apart, the holes aren't plated, and you can't route an isolation slot. That's fine for 24 V, but not for mains. Run the cord through a panel-mount 5×20 fuse holder (0218002.HXP, 2 A slow-blow) and the panel switch straight to the PSB24's L / N / $\perp$ terminals. The board starts at the PSB24's 24 V output.

Board – component side



Each channel runs along one strip, with cuts separating the stages. On strip 2, the DC input goes through PTC1 into D3's clamp, then R2 steps it across a cut onto node A, which runs to CH2. Strip 10 does the same for ohms: PTC2 into node B at D2, then R4 across a cut to node E, which is the 10 V excitation from OUT1. **Every 0 V strip ties to one bare wire along hole 35 on the component side.** Solder it only at the green dots. The copper is scraped off strips 1, 3, 9 and 11 next to the two input segments, so a 480 V mistake on DC+ or Ω has about 4 mm of bare board to jump instead of 1 mm.

Build list

Cuts and prep (copper side)

Cut	strip 2 at hole 6 and hole 13
Cut	strip 10 at hole 6 and hole 13
Cut	strip 20 at hole 30
Cut	strip 30 at hole 18
Scrape	strips 1, 3, 9, 11 from hole 1 to hole 8. Remove all copper, then check with a meter.
Drill	every terminal-block hole to 1.3 mm. The ST2's 0.94 mm holes are too small for the MKDS pins.

Wires

L1	10/12 → 8/12 · node B to the CH3 strip
L2	10/33 → 16/33 · node E to the CH4 strip
L3	20/16 → 30/16 · +24 V to the distribution strip
Bus	bare wire down hole 35, soldered at strips 4, 6, 12, 14, 20, 22 and 30 only

PART	VALUE / PART #	HOLES (STRIP/HOLE)	NOTES
DC VOLTS			
PTC1	TR600-150	2/5 – 2/7	Straddles the cut at 2/6
D3	P6KE39CA	2/9 – 6/9	Stands across strips 3–5. Bidirectional, so either way round.
R2	20 kΩ RN55E2002BB14	2/11 – 2/15	Lies along strip 2, straddling the cut at 2/13
R3	10.7 kΩ RN55E1072BB14	2/18 – 6/18	
D1	P6KE15CA	2/21 – 6/21	
OHMS			
PTC2	TR600-150	10/5 – 10/7	Straddles the cut at 10/6
D2	P6KE15CA	10/9 – 14/9	
R4	2 kΩ RN55E2001BB14	10/11 – 10/15	Straddles the cut at 10/13
AC TRANSDUCER LOOP			
R1	250 Ω RN55E2500BB14	18/6 – 22/6	Stands across strips 19–21

PART	VALUE / PART #	HOLES (STRIP/HOLE)	NOTES
TERMINAL BLOCKS (PHOENIX MKDS 1,5 · 5.08 MM)			
TB4	1715721 (2-pos)	2/2, 4/2	DC+ from F3 · DC COM jack
TB5	1715721 (2-pos)	10/2, 12/2	Ω from F4 · Ω COM jack
TB6	1715721 (2-pos)	18/2, 20/2	VACT (–) · VACT (+)
TB7	1715747 (4-pos)	2, 4, 6, 8 / 37	CH2 · 0V · 0V · CH3
TB8	1715747 (4-pos)	10, 12, 14, 16 / 37	OUT1 · 0V · 0V · CH4
TB9	1715721 (2-pos)	18, 20 / 37	CH1 · 0V
TB10, TB11	1715734 (3-pos) ×2	30 / 4, 6, 8 · 10, 12, 14	+24 V: PSB24 +V · CPU · module · T8C · CPU V1 (output supply) · spare
TB12, TB13	1715734 (3-pos) ×2	30 / 22, 24, 26 · 28, 30, 32	0 V: PSB24 –V · CPU · module · T8C · CPU C0 · buzzer (–)

Terminal blocks needed: 1715721 ×4, 1715747 ×2, 1715734 ×4. That's fewer than the earlier PCB list, since TB1–TB3 and the F5 holder moved off-board.

Before power-up

1. Hold the board up to a light and check every cut and scraped area. No copper bridges, and no solder bridges between strips.
2. Ohmmeter checks with nothing connected: strip 2 (holes 1–5) to anything should read open. Strip 20 (+24) to any 0 V strip should read open. Strip 4 to strip 30 hole 25 should read 0 Ω (bus OK).
3. Single-sided pads lift easily. Support the terminal blocks with a dab of epoxy or hot glue, and hold the block while you tighten screws.
4. Apply 24 V only. Check +24 at TB10 and 0 V at TB12, then follow the same bring-up as the wiring page: DC channel, then ohms (1 k Ω gives about 3.33 V at CH3), then the VACT loop.